AI FOR NURTURING CARE HACKATHON

UNICEF StartUp Lab | MEST Africa | HOPin Academy

GrowWithMe

AI-assisted care reminders, offline risk triage and referral alerts for mothers and children aged 0-59 months in Northern Ghana

Project	GrowWithMe (working title: CareCompanion)
Scope	Caregiver mobile app, Responder app, backend sync platform, on-device AI
Challenge	AI for Nurturing Care Hackathon - maternal and under-five survival in Northern Ghana
Document	Project documentation: overview, problem, solution, users, status, features
Date	October 2026

1. Overview

GrowWithMe is a local-first, AI-assisted health companion for mothers and caregivers in Northern Ghana, covering children aged 0-59 months and maternal health from pregnancy through postnatal care. It combines:

- A personalized care calendar (immunizations, ANC/PNC visits, vitamin A, deworming, growth checks) with SMS and voice delivery for low-literacy, low-connectivity settings.
- An offline, WHO/IMCI- and GHS-aligned symptom and risk assessment that adapts its questions to the situation and classifies results as low, moderate or urgent.
- A location-aware urgent alert flow that connects a family at risk to the nearest verified community health volunteer and health facility by SMS.
- A second companion app for volunteers, nurses and midwives that receives, accepts and closes those alerts - closing the referral loop.
- “Nana”, a caring Ghanaian-elder AI educator powered by an LLM (NVIDIA NIM / Llama 3.1) with a fully offline fallback tier, plus AI meal planning (Nana's Kitchen) and a dietary diversity tracker (Daily Plate).

The system is built as three cooperating parts: a caregiver mobile app (Flutter), a responder mobile app (Flutter) and a Node.js/MongoDB backend that provides sync, SMS (Arkesel), AI, voice (ElevenLabs) and alert routing. Everything in the caregiver app works offline first; connectivity is only needed to sync and to move alerts end to end.

> *Design principle: the app advises and connects - it never diagnoses. Every urgent pathway ends with a human: a volunteer, a nurse or a facility.*

2. The Problem

Maternal, newborn and under-five survival remains a major public health challenge in Northern Ghana (Northern, North East, Savannah, Upper East and Upper West regions), driven by geographic barriers, poverty and weak referral networks.

2.1 Key statistics

Indicator	Reality (2023-2025 data)
Maternal mortality	234 per 100,000 live births (2023) - far above the SDG target of under 70 by 2030
Neonatal / under-five mortality	18 and 36 per 1,000 live births nationally
Neonatal deaths concentrated in the North	Northern Region alone: 10% of all Ghana's neonatal deaths (2,512 deaths, 2019-2023)
Stunting	17.4% nationally, but nearly 30% in Northern and North East regions
Food insecurity	Up to 73.7% in Upper East; over 58% in Savannah
Child food poverty	2.4 million Ghanaian under-fives; only 26.4% of children 6-23 months receive a minimum acceptable diet
Maternal anaemia (Upper West)	Rising, reaching 44.2% among pregnant women

2.2 Why care still fails on the ground

- Working mothers juggle income-generating activities with childcare - immunization, growth monitoring and ANC/PNC visits are easily missed.
- When a child falls ill, caregivers cannot judge how serious symptoms are, and the nearest facility may be far away or costly to reach.
- Referral and follow-up are weak: facilities have limited visibility into who in their catchment is overdue or recovering from a flagged condition.
- CHPS community health volunteers exist but are not systematically connected to households at the moment risk emerges.
- Records are fragmented, connectivity is poor, and most digital health tools assume constant internet and reliable literacy - assumptions that fail in rural Northern Ghana.

2.3 The challenge brief

> *"Develop a practical, low-connectivity, community-level AI solution that empowers frontline health workers and caregivers to accelerate maternal and under-five survival in Northern Ghana."*

3. Proposed Solution

GrowWithMe sits between the caregiver, the community health volunteer and the health facility, and moves information in the direction care needs to flow. It is a three-part system:

3.1 Caregiver app (Flutter, local-first)

- Care calendar generated from the child's date of birth and pregnancy status - Ghana EPI immunizations, vitamin A, deworming, growth checks, ANC/PNC - pushed as notifications, with SMS fallback.
- Offline adaptive triage (child + pregnancy) aligned with WHO IMCI danger signs and GHS protocols; questions adapt to earlier answers.
- Low risk: home-care guidance and watch-for signs. Moderate: nearest CHPS compound/clinic and a scheduled follow-up. Urgent: alert flow triggers.
- Pregnancy journey: LMP/EDD enters an ANC calendar, week-by-week guidance with local baby-size analogies, check-in prompts, and a smooth handoff to postnatal/newborn care at birth.
- Growth monitoring: WHO weight-for-age screening and chart, with offline malnutrition flags.
- Nutrition: age-banded tips using Northern Ghana foods, plus Nana's Kitchen (seasonal, budget-adaptive AI meal plans with offline library fallback) and the Daily Plate 8-food-group tracker.
- Voice-first: questions are read aloud, urgent guidance is always spoken, mic input via on-device speech recognition; home-screen widgets (reminders + "Nana says…" daily briefing).

3.2 Urgent alert flow (closing the referral loop)

On an urgent result the app captures location (with consent) and the backend geo-routes to the nearest available responder within 25 km and facility within 50 km. SMS (Arkesel) goes to all parties; high-risk pregnancies automatically SMS a generated PDF patient report to her registered check-up hospital,

responders and Care Circle. If nothing is nearby, fallback finds the closest hospital via Google Places (name, phone, live-location SMS). Care Circle (father, grandmother) receives SMS copies of all reminders and alerts, because the person who controls transport is often not the mother.

3.3 Responder app (Flutter, second app)

- For volunteers, CHWs, nurses, midwives and doctors: on/off-duty switch, OpenStreetMap with alert and facility pins, alert detail with danger signs, one-tap call, “Navigate” via Google Maps.
- Tiered credentialing (volunteer, CHW, nurse, midwife, doctor) with document upload and admin verification; routing scores tier + verification - distance, so a verified nurse 5 km away beats an unverified volunteer 1 km away.
- Status lifecycle per alert: acknowledged → en_route → at_facility → closed; caregiver receives “help is coming” SMS on acceptance.
- Vitals assessment at the patient's side (all six vitals) using the offline on-device risk model - works in airplane mode.

3.4 Facility visibility and Care Signals

A catchment dashboard API gives CHPS staff: missed visits past grace period, urgent alerts not closed within 24 hours, growth faltering (under -2SD or trending down), third-trimester pregnancies with overdue ANC, and a “lost-to-follow-up radar” ranking families by inactivity and risk. Each signal pairs the dashboard view with a gentle SMS nudge to the caregiver.

3.5 Nana - the AI educator (safety-first architecture)

- Chat + voice assistant with a caring Ghanaian-elder persona, powered by NVIDIA NIM (Llama 3.1 70B) proxied through the backend so keys stay server-side.
- Agentic actions with explicit user confirmation: add a child, start pregnancy tracking, log a weight, read today's plan - executed locally, then synced.
- Hybrid offline: on-device TFLite NLU (17 intents, 5,201 training examples) classifies intents and extracts slots with no connectivity; replies come only from vetted GHS-approved content, so nothing is hallucinated offline.
- Voice: ElevenLabs natural speech via backend proxy with automatic on-device TTS fallback.
- Nana never diagnoses: the deterministic triage engine is the only risk classifier; education and escalation are Nana's job.

4. Target Users

User group	Who they are and what they get
	Working mothers and caregivers of children 0-59 months in Northern Ghana - often with low literacy, irregular connectivity and limited transport money; also track their own pregnancies.
Community health volunteers	CHPS volunteers who receive geo-routed urgent alerts and physically help families reach facilities.

User group	Who they are and what they get
	CHWs, nurses, midwives and doctors using the responder app - with tiered, admin-verified credentials.
Facilities & CHPS staff	Health centres and CHPS compounds that need catchment visibility: missed visits, open urgent cases, growth faltering, lost-to-follow-up.
	Fathers, grandmothers and transport-controlling family members who receive SMS copies of reminders and urgent alerts.
Administrators	Program admins who verify responders, register facilities and manage accounts.

5. State of the Project

GrowWithMe is a working, end-to-end prototype: two Flutter apps, a running backend with real SMS, AI and voice integrations, and a trained on-device ML stack. The core hackathon MVP acceptance criteria (register, add a child, receive reminders, adaptive triage to a risk level, urgent alert to volunteer + facility, facility visibility, and full offline operation of the assessment flow) are all implemented.

5.1 Shipped and working

Area	What exists today
Auth & consent	Phone + OTP auth with real SMS via Arkesel; consent-first onboarding; right-to-erasure endpoint.
Local-first sync	Offline UUIDs, last-write-wins conflict resolution, soft-delete tombstones; sync resilience hardened.
Offline triage	IMCI-aligned adaptive triage for child and pregnancy; deterministic danger-sign rules engine - always the safety floor.
Alerts & referral	Urgent alert flow with geo-routing to nearest volunteer + facility, SMS to all parties; alert PDF reports via Cloudinary.
Care calendar	Ghana EPI immunizations, vitamin A, deworming, growth checks, ANC/PNC; pregnancy journey from LMP/EDD to newborn handoff.
Growth monitoring	WHO weight-for-age screening and charts with offline malnutrition flags.
Nana AI assistant	Chat + voice (NVIDIA NIM Llama 3.1 + ElevenLabs), agentic actions with confirmation, persistent synced conversation memory, quick chips and navigation actions.
Offline AI tiers	On-device TFLite NLU (17 intents, hard eval 23/23) and keyword fallback; device TTS + on-device STT.
Nana's Kitchen + Daily Plate	Seasonal budget-adaptive meal plans and an 8-food-group dietary diversity tracker, both persisted and synced.
Responder app	Second Flutter app: OTP auth, tiered registration with document upload, OSM map, alert lifecycle, deep links, location heartbeat.
Widgets & voice	Reminders widget and Nana daily-briefing home-screen widget; auto-read questions; spoken urgent guidance and first-aid steps.
ML pipeline	Maternal risk TFLite model (~tens of KB, missing-input tolerant); Nana SLM LoRA adapter fine-tuned on 55k synthetic maternal-health conversations; notebooks publish to Hugging Face; model registry with sha256-verified background downloads.

5.2 In progress / next

- Nana SLM inference integration: model infrastructure (download, verification, registry) is ready; wiring an on-device inference engine (llama.cpp via FFI / MediaPipe) is the remaining step.
- Facility dashboard web UI (the API already exists).
- Care Signals scan hardening and scheduling polish.
- Recorded Dagbani/local-language audio prompts through the voice pipeline.

- SMS/USSD companion channel for feature phones; immunization card photo backup.
- Collected local case data to fine-tune and localize the risk model to Tamale (Phase 4 plan is documented).

5.3 Honest limitations

- No FCM push yet - SMS is the push channel by design for now.
- Responder verification is manual by an admin (documented, pragmatic for the current stage).
- The SLM (Llama 3.2 adapter) cannot run inference on-device yet; the 3-tier fallback (LLM → NLU → keywords) covers offline behavior meanwhile.
- Local-language speech is English-first today; Dagbani audio prompts are on the roadmap.

6. Features at a Glance

Feature	Caregiver	Responder	Offline
Phone OTP + consent onboarding	Yes	Yes	SMS needed once
Care calendar & reminders (EPI, ANC/PNC, vit A, deworming)	Yes	-	Yes (SMS fallback)
IMCI-aligned adaptive triage	Yes	-	Fully offline
Urgent alerts: geo-routing + SMS loop	Yes	Receives	Queued until connectivity
Pregnancy journey + AI check-ins	Yes	-	Core offline
Growth monitoring (WHO charts)	Yes	-	Fully offline
Nana AI educator (chat + voice)	Yes	-	NLU + keywords offline
Nana's Kitchen meal plans / Daily Plate	Yes	-	Offline library fallback
Vitals risk scoring (ML, missing-input tolerant)	Basic	Assess vitals	Fully offline
Alert lifecycle + navigation	-	Yes	Map works offline
Care Circle SMS copies	Yes	-	-
Catchment dashboard / Care Signals	-	Yes (API)	Backend-side

> *“The AI works where the network doesn't.” Demo story: a nurse accepts an alert in Tamale with no data connection, enters BP 145/95, blood sugar 8.2, temperature 99.5 - the phone, in airplane mode, returns “HIGH risk - 84%” in under 50 ms and she refers immediately. When signal returns, the assessment syncs into the case timeline automatically.*

7. Additional Information

7.1 Technology stack

Layer	Details
Mobile apps	Flutter (Dart, Riverpod, Drift local DB, dio, flutter_local_notifications, home_widget, fl_chart, flutter_tts, speech_to_text, tflite_flutter)
Backend	Node.js, Express, MongoDB (Mongoose), JWT access+refresh auth, node-cron daily reminder job
Integrations	Arkesel (OTP + alert + reminder SMS), Cloudinary (uploads + model/PDF hosting), NVIDIA NIM LLM (Llama 3.1 70B), ElevenLabs TTS, Google Places fallback
ML & data	Google Colab training notebooks; UCI Maternal Health Risk dataset (1,014 rows); 55k-example synthetic Nana conversation dataset; NLU v2 (17 intents, 5,201 examples) - Hugging Face distribution
Maps	flutter_map / OpenStreetMap (no API key) in the responder app; Google Maps only for turn-by-turn deep links

7.2 Offline AI architecture

Three tiers with a deterministic safety floor: (1) backend LLM online; (2) on-device TFLite NLU with curated responses offline; (3) keyword matching as the ultimate fallback. Downcoming tiers: a maternal-risk vitals model (~tens of KB, accepts any subset of six vitals) and an optional “Nana Lite” on-device LLM (~500 MB, Wi-Fi-only download). In every configuration, the deterministic danger-sign rules engine runs in front of any AI and can raise a verdict, never lower it - urgent stays urgent.

7.3 Safety, privacy and responsibility

- The app advises and connects; it never diagnoses. Every urgent pathway ends with a human.
- All triage thresholds err on the side of escalation; ambiguous cases are treated as higher risk.
- Consent flags captured at onboarding (data processing, location on urgent, SMS reminders) - aligned with Ghana's Data Protection Act and the MEST privacy policy.
- Location is stored only on urgent assessments/alerts, never tracked continuously.
- AI responses always carry an “AI can make mistakes” disclaimer; Nana is constrained to GHS/WHO-vetted content when offline.
- Right to erasure: DELETE /users/me removes the account and every owned record.

7.4 Repository & structure

Path	Contents
mobile/	Caregiver Flutter app (the primary app)
responder/	Responder Flutter app (alerts, map, credentialing)
backend/	Node/Express/MongoDB sync + AI + SMS/routing platform
ml/	Training notebooks, model artifacts and datasets
doc/, *.md	Concept notes, roadmap, offline model plan, responder plan, SLM integration guides

GrowWithMe - practical, responsible AI for nurturing care in Northern Ghana. Built to empower health workers and caregivers, not to replace them.